Information Engineering

1-Introduction To Information Engineering Concepts. “IE —INT 461

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Sinai University

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Outline

1. Course Logistics
2. Course Polices
3. Background
4. Introduction to Information Engineering
5. IE-Taxonomy
6. IE Terminology
7. A Framework for System Analysis and Design
8. The players in Information Game “the System stakholders”

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General Information

- Instructor
  - Nasser M. Tamim
- Location
  - N-2-24
- Class Time
  - Every Tuesday at 9 am: 10.30 am
- Email
  - Nasser.Tamim@su.edu.eg
- Office Hours
  - Tuesday 10.40: 11.40 pm

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Course Literature

- Prerequisites
  - INT 232 Computer Networks, CSW263 Software Engineering)
- Credit Hours
  - (2+0+2).
- Text Book(s)
  - "Bentlety and Whitten, (2007). System Analysis and Design Methods, 7th Edition, McGraw-Hill".
- References
  - " Jeffrey A., V. Ramesh, and Heikki Topi, (2016). Modern Database Management, 13 Edition, Pearson".
  - " Clive Finkelstein (1980). Information Engineering Methodology ".
- Course Website

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Course Web Site

The LMS for the course page contains:

1. Lecture slides
2. Lecture Videos if any
3. Handouts
4. Exercises/Assignments.

Course Objectives

This course will help students to achieve the following objectives:

1. To understand, what information engineering is, and why it is important.
2. To differentiated between the concept of Information Engineering and software Engineering and Information technology.
3. To understand the ethical and professional issues concern to Information engineers.
4. tem to highlights the role Information Engineering in business process in building different type of Information systems.

Course Outline

The following is an outline for the course, subject to alteration as the semester proceeds.

Week	topic	Value
1	Concept of Information Engineering and Information Engineers	
2	Information Architecture	
3	Information Engineering Project	
4	Information Strategic Planning	
5	development of strategic planning	
6	Development of strategic Models	
7	Business Process	
8	Re-Engineering	
9	IT- Security and Quality and IT quality Standers	
10	Information plans	
11	Information Distribution and integrity	
12	National and International Information	

Table: Caption

Team Projects

The following are some proposed team-project that will be implemented and documented in the Lab:

1. Parking Garage Automation:
   - Develops a computerized system to manage parking usage and online reservations for a parking garage.
2. Traffic Monitoring:
   - Develops a system to inform the user of road and traffic conditions in the area of interest.
3. Restaurant Automation:
   - Develops a computerized system to help restaurant personnel co-ordinate their activities and improve their services, and for the management to track business growth and create future plans.
4. Khan el-Khalili Bazar
   - Develops an online computerized system to introduce the Egyptian civilization globally.
5. More case studies can be seen at `https://software-engineering-book.com/case-studies/`

Laboratory Activities

Lab activity will be use software engineering CASE tools such as:

1. Draw.io
2. PERF
3. Star UML, Microsoft Word
4. Microsoft Project, Oracle or MS SQL or MS ACCESS.
   Student must read / do Post-Lab before come to the laboratory. Lab will be a lab assessment/lab team-project.

Assessment Activities

Students will be assessed using exams, quizzes, homework assignments and/or project work typically distribution as follows:

Evaluation item	Percentage
Assignments	5%
Lab	15%
Mid-Term Exam	20%
Final Exam	60%

Exam Policy

- Exams will be a combination of the materials presented in the lectures and homework problems.
- Missed tests **Cannot be made up**
  - Missed tests **Unfortunately cannot be made up**—there will be no makeup exams; students must instead make arrangements with the instructor to take the exam prior to the planned absence.
  - **Consequently,** no makeup for missed quizzes.

Attendance/Absence Policy

- The University rulers requires that the attendance is recorded. **Therefore** the Class attendance will be taken in every lecture.
- Attend all classes Experience shows;
  - **Poor attendance = Poor performance**.
    So, attend all classes.The variability of class lecture slides/videos is no substitute for class attendance; Do not banned me to see you Always.
    ☺☺☺☺☺☺☺☺☺☺☺☺☺☺☺☺☺☺☹
- **WARNING**
  - Students that exceed class absences 25% of lectures are being considered dropped from the course and prevented from entering the final exam.

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Academic Dishonesty

TRUST, TRUST Nothing else BUT TRUST

- The student-teacher relationship is based on trust.
  - Your name on the submitted work is an affirmation that the work is entirely your own.
  - When submitting team work, all team members' names must appear on the work.
  - Any student engaging in academic dishonesty, or assisting another student in academic dishonesty.
    - Will gain a grade of **F** , " Wel2yazo Bel ALLAH !!!! " for the course.
    - Reported to Administration.
- Academic Dishonesty
  - Copying another student's work, take care I use plagiarism software!!!!!!!!!!!!!!!!!!!
  - Submitting the work of another as your own.
  - Allowing others to copy your work.

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Information/ Engineering Gold / Mining



In today's digital world, information is one of the most valuable goods in the economy. Information is important in scientific investigation, with a potentially enormous economic, industrial, and social impact. In addition, it contributes to strategic organizational development.
A variety of critical systems heavily depend on information. Such systems include

- Health care systems, banking/ATM, computing, geographical information system (GIS),biotechnology
- air traffic control
- hotel/airline reservation

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Information Engineering versus Information Technology

1. Information Engineering and Information Technology
   - Information Engineering is the engineering discipline that deals with the generation, distribution, design, analysis, and use of data, information, and knowledge in systems.
   - **or** communication.
     - So It is data driven approach.
   - Information Technology (IT)
     - It is a broad subject encompassing all aspect of IT field; Computer Hardware, Computer Software, Tele-communication, etc..
2. So, Information Engineering is an approach to planning, generating, distributing, analyzing, and using collections of data in systems to facilitate decision making and business communication and requirements

Information Engineering (IE) roots

- IE is an approach to **designing and developing** information systems.
- IE has a somewhat chequered history that follows two very distinct threads.
- It is said to have originated in Australia between 1976 and 1980,
- It appears first in the literature in 1981 in the**Savant Institute publication Information Engineering** by **James Martin** and **Clive Finkelstein**
- **Clive Frankenstein** is regarded as the father of information engineering because he wrote the first publication on information engineering.

Information Engineering (IE) roots

Information Engineering

Firsty provided data analysis and database design techniques that could be used by database administrators (DBAs) and by systems analysts to develop database designs and systems based upon an understanding of the operational processing needs of organizations for the 1980s

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Information engineering - Overview

In software engineering

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Information engineering (IE) or information engineering methodology (IEM) is an approach to designing and developing information systems.

It can also be considered as the generation, distribution, analysis and use of information in systems.

The different type of Information engineering“IE"

There are two type of Information Engineering.

- Data processes Driven Variant of Information Engineering **DP-Information Engineering**
- Business processes Driven Variant of Information Engineering **BP-Information Engineering**

The components of information engineering include more theoretical fields such as:

- Machine Learning (ML),
- Control Theory
- Information Theory
- Biomedical Systems
- Signal Processing
- Image Processing

The different type of Information engineering“IE"

- Cheminformatics
- Bioinformatices in which information engineering tackles the high-throughput biological data for analysis and better biological understanding.

Many of these originate from computer science, as well as other branches of engineering such as computer engineering, electrical engineering, and bio-engineering.

DP-Information Engineering

The first variant was developed in the USA by Database Design. They changed the data-driven emphasis of IE from 1976-1980 instead to a process-driven focus for use by DBAs, systems analysts and Data Administrators (DAs).
These are Data Processing (DP) staff roles; people in these roles generally take a DP-driven focus on the Four systems development phases were defined as follows:

- Planning Phase
- Analysis phase
- Designing Phase
- Implementing OR Constructing phase

The Disadvantage of DP-Driven of IE

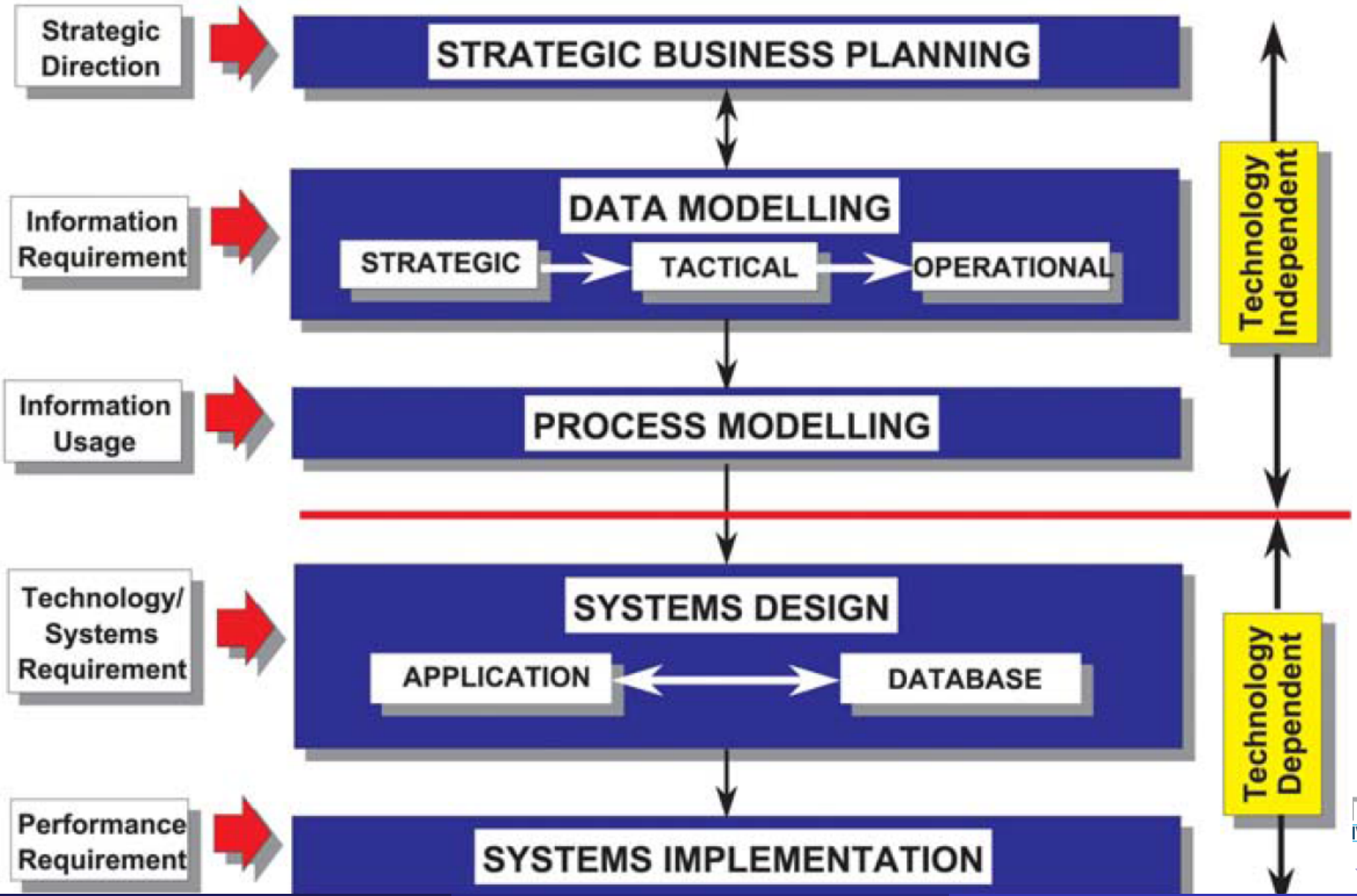
- use of third and fourth generation languages to develop systems and databases with the DP-driven variant of IE resulted in long development times.
- The resulting systems are too volatile. Because Business changes could not easily be anticipated, as these changes often occur without warning at the operational levels of organizations.

Business-Driven Information Engineering 1

Business-driven Information Engineering uses methods that are understandable and applied by business managers and their staff (the business experts) as well as by IT staff.
It uses a number of phases to capture the business knowledge and understanding vital for success, summarized below and illustrated in the following:

- Strategic Planning uses the strategic directions set by management to identify their information needs.
- Data Modelling documents in data models the information and data needed to achieve those directions.
- Process Modelling defines the business processes based on information usage, to implement the plans.

Business-Driven Information Engineering 2



The formal definition of information Engineering

- **James Martin (author)** Information Engineering first provided data analysis and database design techniques that could be used by database administrators (DBAs) and by (System Analyst) to develop database designs and systems based upon an understanding of the operational processing needs of organizations.
- It assumes logical data representations are relatively stable, as opposed to the frequently changing processes that use the data.
  Therefore, the logical data model, which reflects an organization's rules and policies, should be the basis for systems development

The Information Engineering (IE) methodology's

- focus is on the data of the organization. First, the data are represented by a data model such as:
  - First, An Entity Relationship Diagram "ERD".
  - Second, the processes are represented by another model such as an Activity Hierarchy Diagram.
  - or Data flow diagram, DFD.
- The information engineering methodology supports an assumption that data should be the primary focus of a development methodology while traditional methodologies support processes as the primary focus.

IE as a interlocking set of formal techniques

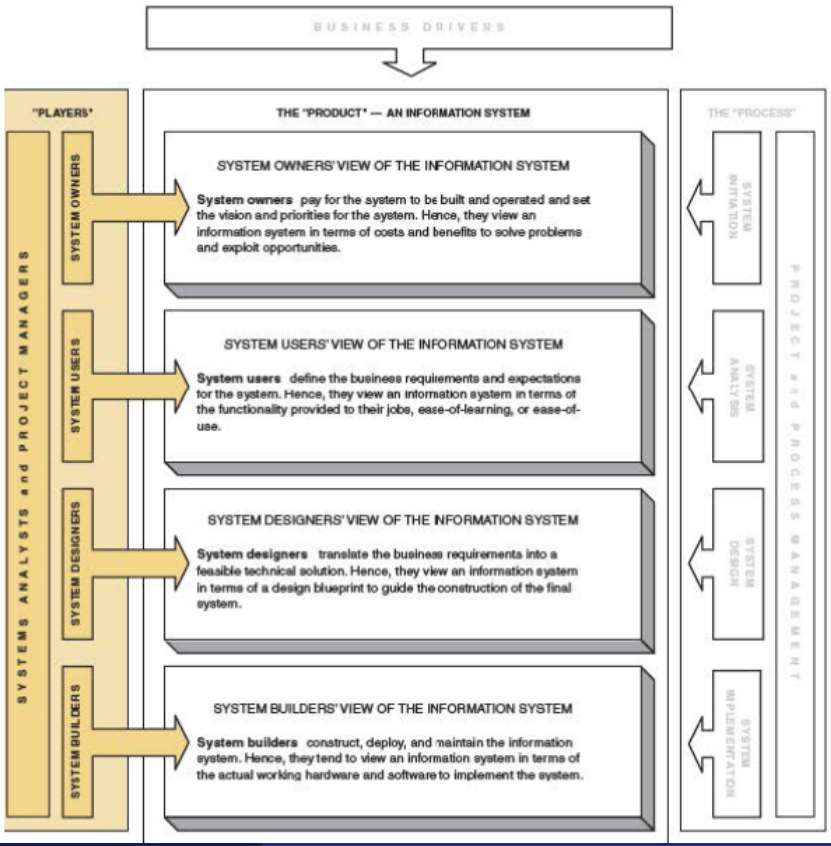
- IE is an interlocking set of formal techniques in which enterprise models, data models and process models are built and are used to create and maintain data processing systems James Martin (1986)

TERMINOLOGY

- **A system** is a group of interrelated components that function together to achieve a desired result For instance, you may own a home theater system made up of a DVD player, receiver, speakers, and display monitor.
- **Information systems (IS)** in 0organizations capture and manage data to produce useful information that supports an organization and its employees, customers, suppliers, and partners.
- **Transaction Processing System (TPSs)**
  process business transactions such as orders, time cards, payments, and reservations.
- **Management information systems (MISs.)** use the transaction data to produce information needed by managers to run the business.
- **Decision support systems (DSS)**
- **Executive information systems (EISs)**
- **Expert systems (ESs)**
- **Communication and collaboration systems**
- **Office Automation systems**

IT Stakeholders

Stakeholders any person who has an interest in an existing or proposed information system and it may include both technical and non technical workers in information industry and can extended to the external and internal



IT Stakeholders Rule

- Systems Owners is the person who pay to build and operate and set the vision and priorities for the system, hence they view an information system as in term of cost and benefits to solve problems and exploit opportunities.
- system users; define the business requirements and expectation for the system. Hence they view information system as in terms of the functionality provided to their Jobs ease-of-learning or ease-of-use.
- System Designer
- system builders.
- System analyst (Information Engineer)

System Analyst and its rule

Their is a huge gap between (system owner, user, designer and builder views; and both of them has a different perspectives on the system. While some has generally view the other has a details views. some are nontechnical the other are is.
So we need a bridge that gap; the person who act this bridge is call —System Analyst or one of the following synonyms: System architect, OR systems engineer OR information engineer, OR information analyst, OR systems information.

The skills of the system analyst

There is no perfect tools or techniques, process or principle or methodology all the time but only Basic concepts and principle of system thinking
The role of the Systems Analyst Systems analyst understand both business and computing,business, management.etc...

where is a system analyst work??????????

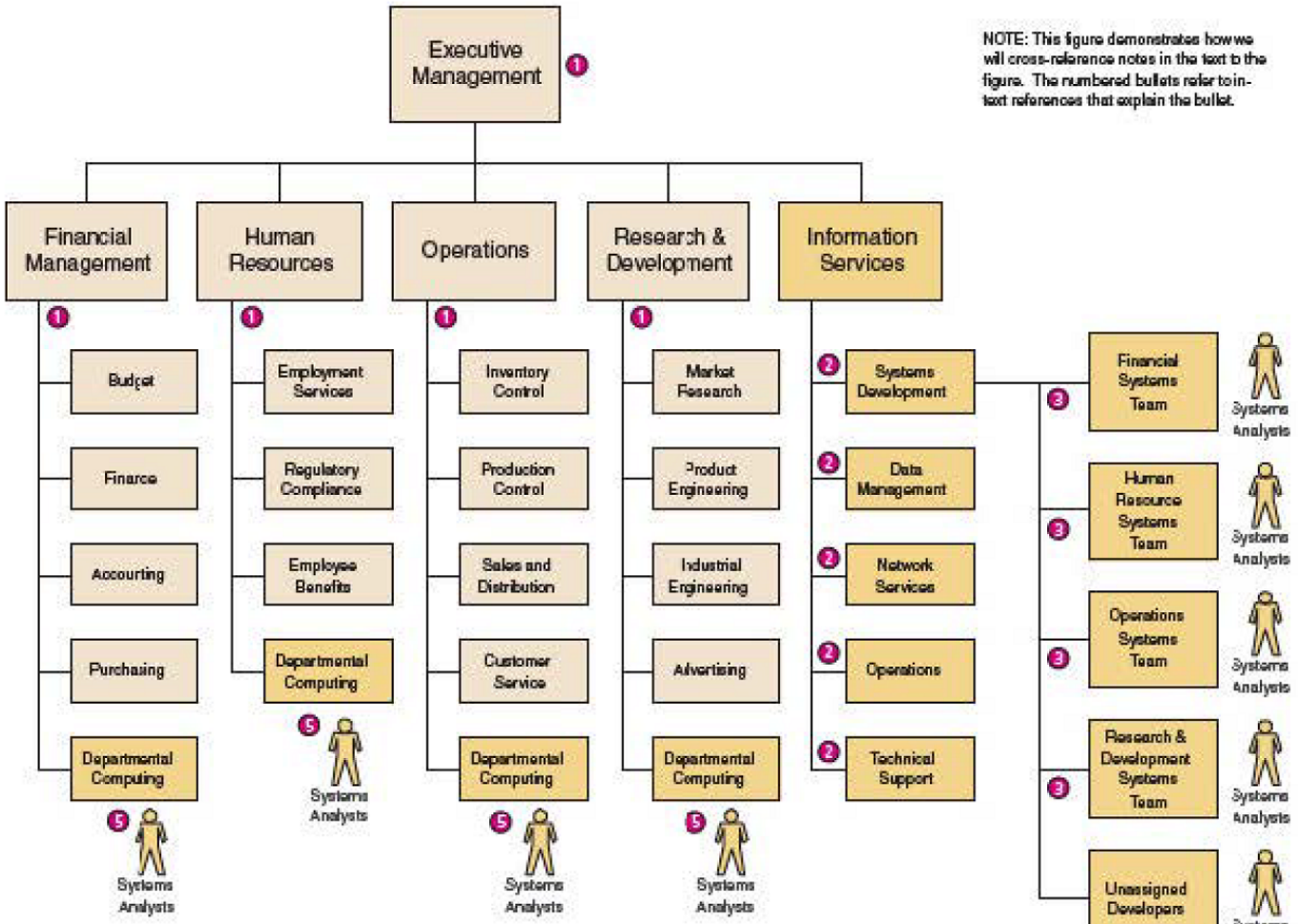


FIGURE 1-2 Systems Analysts in a Typical Organization

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Skills Needed by the Systems Analyst

1. Working knowledge of information technologies (knowing every new)
2. Computer programming experience and expertise.
3. General knowledge of business processes and its terminology.
4. General problem solver skilles.
5. Good interpersonal communication skills.
6. good interpersonal relations skills.
7. flexibility and adaptability.
8. character and ethics.

Frame Title

The Context of Systems Development Projects

FIGURE 1-4 Ethics for Systems Analysts

The Ten Commandments of Computer Ethics

1. Thou shalt not use a computer to harm other people.
2. Thou shalt not interfere with other people's computer work.
3. Thou shalt not snoop around in other people's computer files.
4. Thou shalt not use a computer to steal.
5. Thou shalt not use a computer to bear false witness.
6. Thou shalt not copy or use proprietary software for which you have not paid.
7. Thou shalt not use other people's computer resources without authorization or proper compensation.
8. Thou shalt not appropriate other people's intellectual output.
9. Thou shalt think about the social consequences of the program you are writing or the system you are designing.
10. Thou shalt always use a computer in ways that insure consideration and respect for your fellow humans.

Source: Computer Ethics Institute

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Major phases of Information Engineering

- Scope definition
- Problem Analysis
- Requirement Analysis
- Logical Design
- Decision Analysis
- physical design and integration
- Construction and testing
- Installation and delivery